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Sub :- Computer vision

Code :- ITA - 0501

· CO2 - AT.3

CLASS TEST.

1) Gamma Transformation :-

↳ used to brighten dark images.

i) Read the dark image.
ii) Apply gamma correction with gamma less than 1.
iii) Increase the brightness.
iv) Display the image.

Result :- The dark image becomes brighter.

2) Histogram Equalization :-

↳ improves the contrast of an image.

i) Convert image to grayscale.
ii) calculate pixel intensity values.
iii) calculate cumulative distribution.
iv) Spread the intensity values.
v) Display the new image.

Result :- fingerprint ridges becomes clearer.

3) median filter :-

↳ removes black and white noise.

i) Select a 3x3 pixel window.
ii) Arrange pixel value in order.
iii) find the middle value.
iv) Replace the center pixel with it.

4) **unsharp Masking :-**

↳ It makes blurred images sharper.

i) Read the orginal image.
ii) create a blurred image.
iii) subtract the blurred image from the original.
iv) Add the difference to the original.
v) Display the result.

Result :- circuit board tracks become sharper.

5) **· Low-pass filter :-**

↳ Removes unwanted noise from an image.

i) Convert the image to the frequency domain.
ii) Apply a Gaussian low-pass filter.
iii) Remove high-frequency noise.
iv) convert the image back.
v) Display the image.

Result :- The image becomes smoother.

6) **High-Boost filter :-**

↳ Improves fine details in retinal images.

i) Read the retinal image.
ii) create a blurred image.

iii) Extract the details.

iv) Add the details to the original image.

v) Display the result.

Result:- Blood vessels become clearer.

7) Point - and Line Detection:-

↳ Detects air bubbles and thin cracks.

i) convert the image to grayscale.

ii) use a point mask to detect bubbles.

iii) use line masks to detect cracks.

iv) Apply a threshold to identify defects.

v) mark the detected defects.

Result:- Bubbles and cracks are identified.

8) Adaptive thresholding:-

↳ It separates hard writing from the background.

i) Convert the image to grayscale.

ii) Divide the image into small regions.

iii) calculate a threshold for each region.

iv) convert ink to black and the background to white.

v) Display the result.

Result:- Handwriting becomes clear even under uneven lighting.

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9) Region Growing:-

↳ Separates a lake from surrounding land.

i) select a starting point inside the lake.

ii) check neighbouring pixels.

iii) compare their intensity values.

iv) Add similar pixels to the region.

v) Repeat until no similar pixels remain.

Result:- The lake is separated from the land.

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10) Edge linking:-

↳ connects broken edges of a gear.

i) Detects the gear edges.

ii) Find the broken edges.

iii) connect nearby edges.

iv) use the Hough Transform to identify circular patterns.

v) obtain the complete boundary.

Result:- The gear outline becomes clearer and more complete.